One adjacent swap defeats a pairwise readout

Only the AB relation changes; a conventional XOR bit supplies extra information

History	PAIRWISE COMPARISONS AB	AC	AD	BC	BD	CD	EXTRA BIT XOR
A B C D	1	1	1	1	1	1	0
B A C D	0	1	1	1	1	1	1

AB erased — The six pairwise entries are now identical

A B C D	?	1	1	1	1	1	0
B A C D	?	1	1	1	1	1	1

Pairwise record: {ABCD, BACD} **+ XOR: unique recovery**

Known erasure, four single-use events. An added coded bit is not a demonstrated mechanical part. One parity bit detects one wrong bit; it cannot generally correct that unmarked error.